

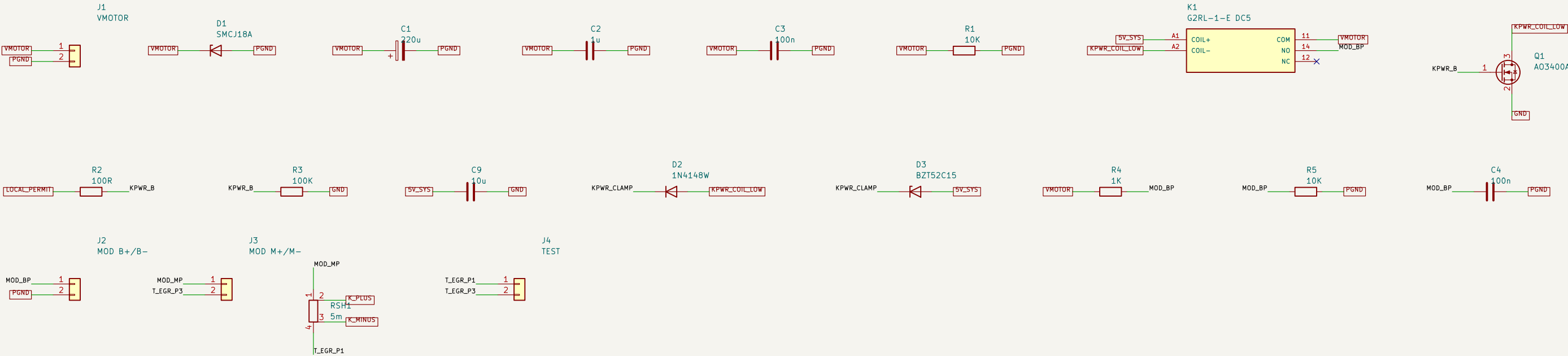


P07-S1 / 02 VMOTOR, KPWR, MODUL B+/B-, BOCZNIK, PORT TEST

VMOTOR (P02 R4 J2, za F1 MINI 7,5 A) -> J1 -> KPWR K1 (NO) -> MOD\_BP -> J2 -> moduł B+. R4 1k ładuje 330 uF modulu przy otwartym KPWR; KPWR: cewka 5 V z 5V\_SYS (decyzja 5.10). Q1 AO3400A (zrodło na GND) z LOCAL\_PERMIT; clamp D2 + D3 15 V do 5V\_SYS.

Modul M+ -> J3.1 -> RSH1 5 mOhm (Kelvin K\_PLUS/K\_MINUS) -> T\_EGR\_P1 -> J4.1 -> port TEST (P11). M- -> J3.2 = T\_EGR\_P3 -> J4.2. Prad dodatni: kierunek A (RPWM, M+ wyżej).

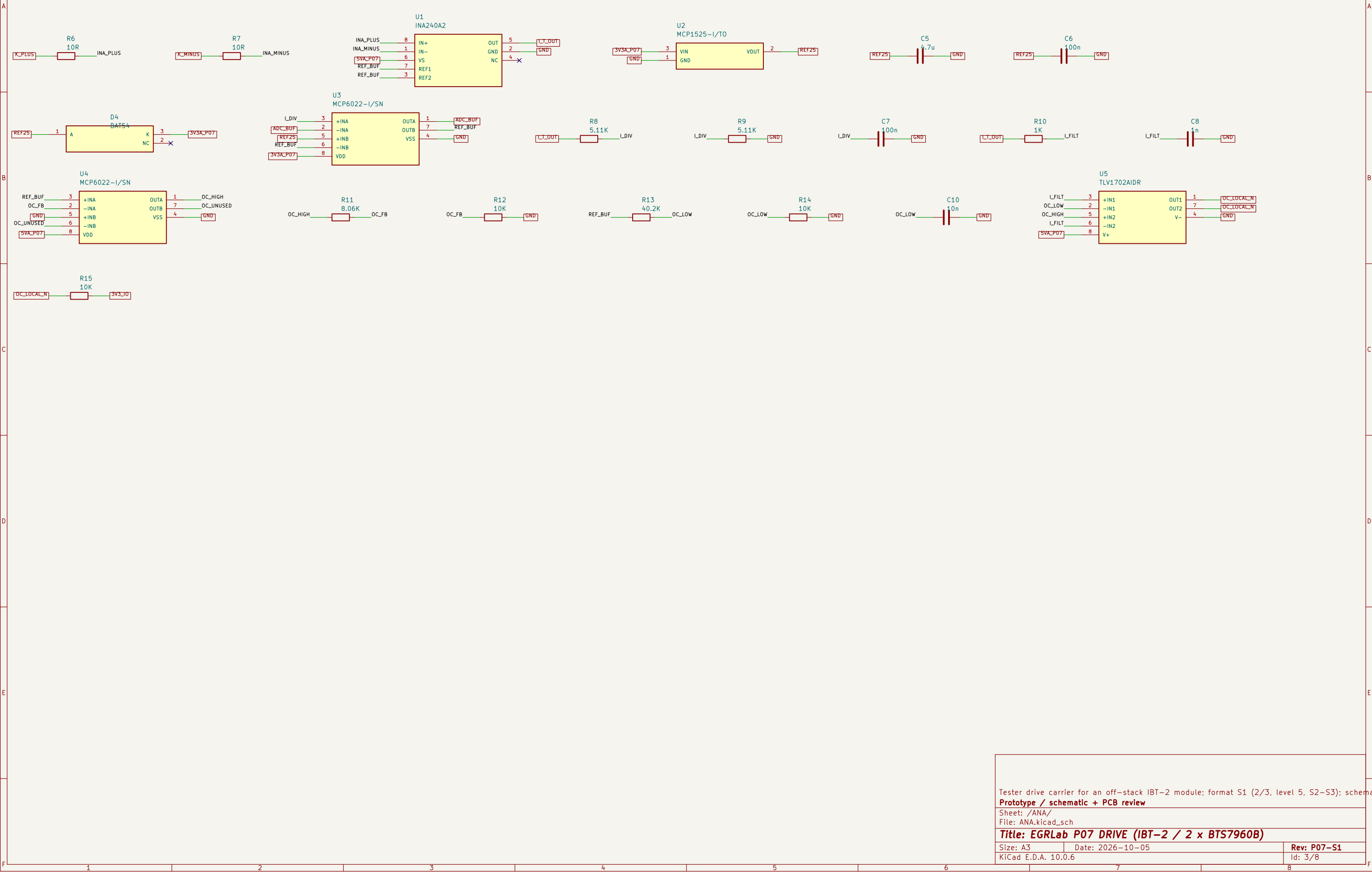
PGND: powrot przez J1.2 do P02, B- modulu przez J2.2, TVS D1, C1-C4, R1/R5 (obwód cewki KPWR w domenie GND). Tor 10 A: pola >= 4 mm na obu warstwach (S1 3, jak P06 R2).



P07-S1 / 03    INA240, ODNIESIENIE, ITEST, OKNO OC

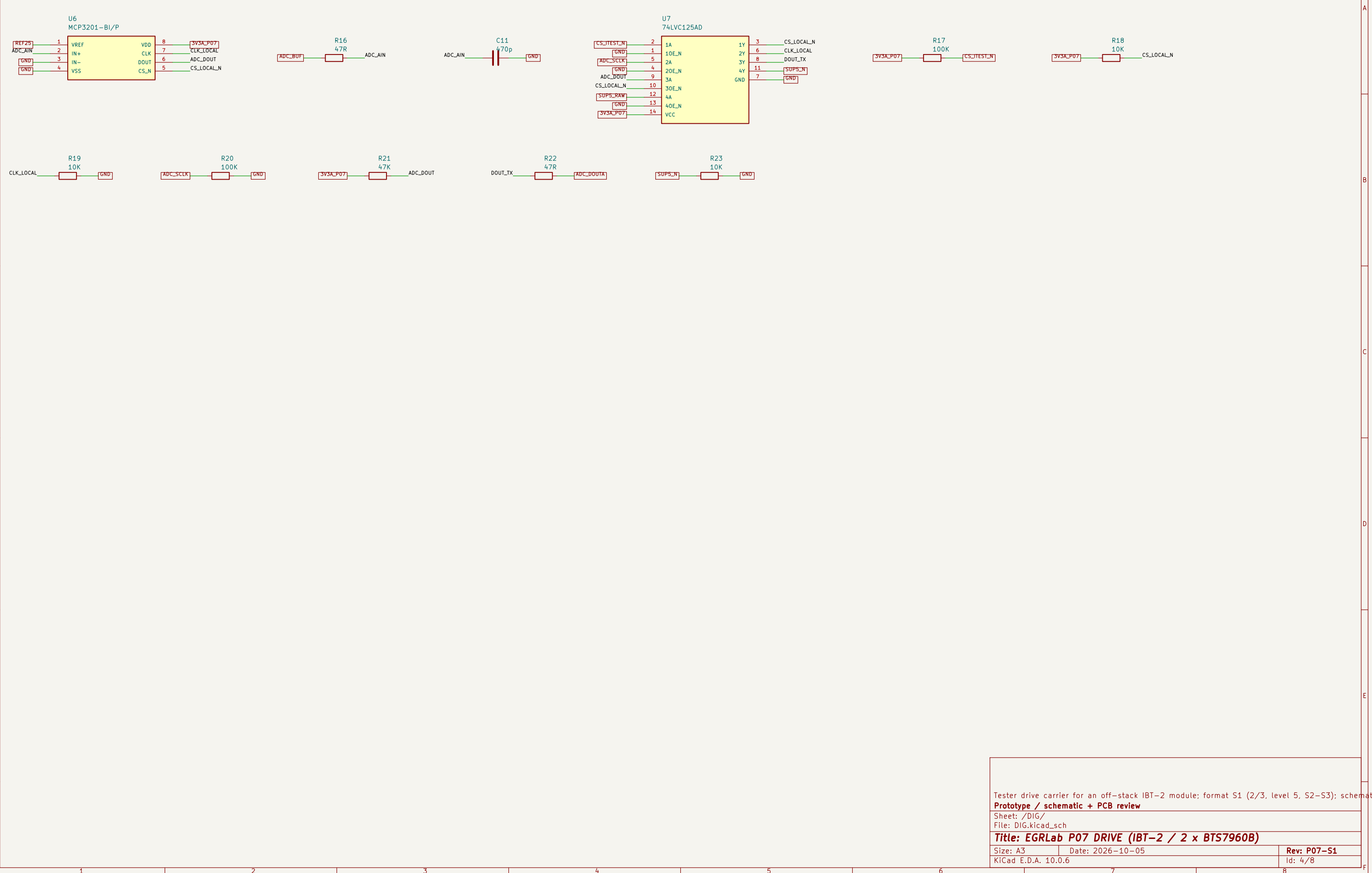
$L_T_{OUT} = REF\_BUF + 50 \times 5 \text{ mOhm} \times I = 2,5 \text{ V} + 0,25 \text{ V/A}$ . ITEST: R8/R9 1:2 + C7 100n (tau 0,26 ms) -> U3A -> MCP3201: 1,25 V + 0,125 V/A, +-10 A w 0..2,5 V.

Okno OC: L\_FILT (R10 1k / C8 1n) vs OC\_HIGH = REF\_BUF x 1,806 (4,515 V, +8,06 A) i OC\_LOW = REF\_BUF x 0,1992 (0,498 V, -8,01 A), U5 open collector -> OC\_LOCAL\_N (R15 10k do 3V3\_IO).



P07-S1 / 04 MCP3201 ITEST I BUFOR SPI

SPI mode 0, 16 zegarów, raw = (word >> 1) & 0x0FFF (jak P06 R2). DOUT\_TX trojstanowy (OE = CS\_LOCAL\_N) + R22 47R na wspólnej ADC\_DOUTA. U7 gate 4: SUP5\_RAW (5 V) -> SUP5\_N (3,3 V).

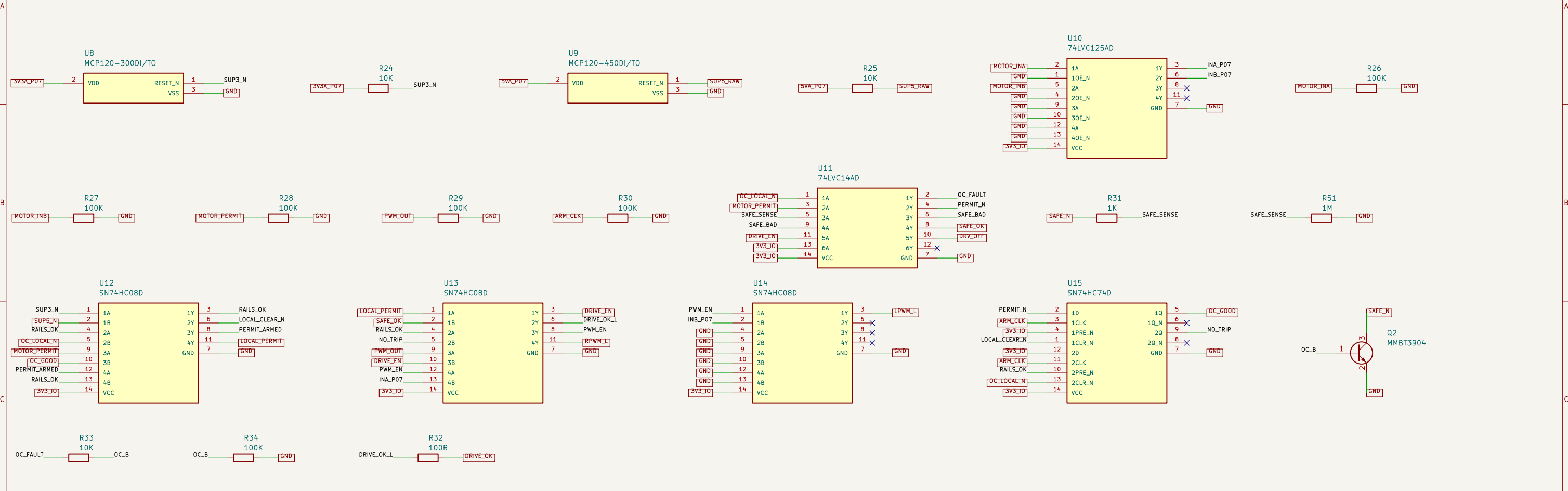


|                                                               |   |   |   |   |   |   |   |
|---------------------------------------------------------------|---|---|---|---|---|---|---|
| 1                                                             | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
| P07-S1 / 05 NADZORCY, ODBIORNIKI, BRAMKI, ZATRZASK OC, SAFE_N |   |   |   |   |   |   |   |

RAILS\_OK = SUP3\_N & SUP5\_N. LOCAL\_CLEAR\_N = RAILS\_OK & OC\_LOCAL\_N. OC\_GOOD: kasowany przez OC/zasilanie, ustawiany TYLKO zboczem ARM\_CLK przy MOTOR\_PERMIT = L (D = PERMIT\_N).

LOCAL\_PERMIT = MOTOR\_PERMIT & OC\_GOOD & RAILS\_OK (-> KPWR), DRIVE\_EN = LOCAL\_PERMIT & SAFE\_OK, RPWM = PWM\_OUT & DRIVE\_EN & MOTOR\_JNA; LPWM = PWM\_OUT & DRIVE\_EN & MOTOR\_INB.

NO\_TRIP (FF2): PRE = RAILS\_OK (start = brak zadzialania), CLR = OC\_LOCAL\_N, D = 1 przy ARM\_CLK. DRIVE\_OK = RAILS\_OK & NO\_TRIP. Q2 sciaga SAFE\_N podczas OC.

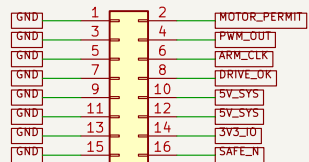
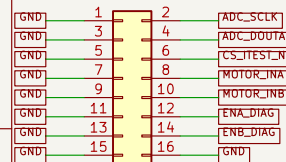




P07-S1 / 07 J\_BP1 / J\_BP2 DO P12 (KRAWEDZ A)

J\_BP1 (S2, x płytki 26,5 / stosu 80,0): ADC\_SCLK 2 / ADC\_DOUTA 4 jak J\_BP2 P03 R6, P05 R3 i J\_BP P06 R2; CS\_ITEST\_N, MOTOR\_INA/INB, ENA/ENB\_DIAG do P03 R6.

J\_BP2 (S3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232, 233, 234, 235, 236, 237, 238, 239, 240, 241, 242, 243, 244, 245, 246, 247, 248, 249, 250, 251, 252, 253, 254, 255, 256, 257, 258, 259, 260, 261, 262, 263, 264, 265, 266, 267, 268, 269, 270, 271, 272, 273, 274, 275, 276, 277, 278, 279, 280, 281, 282, 283, 284, 285, 286, 287, 288, 289, 290, 291, 292, 293, 294, 295, 296, 297, 298, 299, 300, 301, 302, 303, 304, 305, 306, 307, 308, 309, 310, 311, 312, 313, 314, 315, 316, 317, 318, 319, 320, 321, 322, 323, 324, 325, 326, 327, 328, 329, 330, 331, 332, 333, 334, 335, 336, 337, 338, 339, 340, 341, 342, 343, 344, 345, 346, 347, 348, 349, 350, 351, 352, 353, 354, 355, 356, 357, 358, 359, 360, 361, 362, 363, 364, 365, 366, 367, 368, 369, 370, 371, 372, 373, 374, 375, 376, 377, 378, 379, 380, 381, 382, 383, 384, 385, 386, 387, 388, 389, 390, 391, 392, 393, 394, 395, 396, 397, 398, 399, 400, 401, 402, 403, 404, 405, 406, 407, 408, 409, 410, 411, 412, 413, 414, 415, 416, 417, 418, 419, 420, 421, 422, 423, 424, 425, 426, 427, 428, 429, 430, 431, 432, 433, 434, 435, 436, 437, 438, 439, 440, 441, 442, 443, 444, 445, 446, 447, 448, 449, 450, 451, 452, 453, 454, 455, 456, 457, 458, 459, 460, 461, 462, 463, 464, 465, 466, 467, 468, 469, 470, 471, 472, 473, 474, 475, 476, 477, 478, 479, 480, 481, 482, 483, 484, 485, 486, 487, 488, 489, 490, 491, 492, 493, 494, 495, 496, 497, 498, 499, 500, 501, 502, 503, 504, 505, 506, 507, 508, 509, 510, 511, 512, 513, 514, 515, 516, 517, 518, 519, 520, 521, 522, 523, 524, 525, 526, 527, 528, 529, 530, 531, 532, 533, 534, 535, 536, 537, 538, 539, 540, 541, 542, 543, 544, 545, 546, 547, 548, 549, 550, 551, 552, 553, 554, 555, 556, 557, 558, 559, 560, 561, 562, 563, 564, 565, 566, 567, 568, 569, 570, 571, 572, 573, 574, 575, 576, 577, 578, 579, 580, 581, 582, 583, 584, 585, 586, 587, 588, 589, 590, 591, 592, 593, 594, 595, 596, 597, 598, 599, 600, 601, 602, 603, 604, 605, 606, 607, 608, 609, 610, 611, 612, 613, 614, 615, 616, 617, 618, 619, 620, 621, 622, 623, 624, 625, 626, 627, 628, 629, 630, 631, 632, 633, 634, 635, 636, 637, 638, 639, 640, 641, 642, 643, 644, 645, 646, 647, 648, 649, 650, 651, 652, 653, 654, 655, 656, 657, 658, 659, 660, 661, 662, 663, 664, 665, 666, 667, 668, 669, 670, 671, 672, 673, 674, 675, 676, 677, 678, 679, 680, 681, 682, 683, 684, 685, 686, 687, 688, 689, 690, 691, 692, 693, 694, 695, 696, 697, 698, 699, 700, 701, 702, 703, 704, 705, 706, 707, 708, 709, 710, 711, 712, 713, 714, 715, 716, 717, 718, 719, 720, 721, 722, 723, 724, 725, 726, 727, 728, 729, 730, 731, 732, 733, 734, 735, 736, 737, 738, 739, 740, 741, 742, 743, 744, 745, 746, 747, 748, 749, 750, 751, 752, 753, 754, 755, 756, 757, 758, 759, 760, 761, 762, 763, 764, 765, 766, 767, 768, 769, 770, 771, 772, 773, 774, 775, 776, 777, 778, 779, 780, 781, 782, 783, 784, 785, 786, 787, 788, 789, 790, 791, 792, 793, 794, 795, 796, 797, 798, 799, 800, 801, 802, 803, 804, 805, 806, 807, 808, 809, 810, 811, 812, 813, 814, 815, 816, 817, 818, 819, 820, 821, 822, 823, 824, 825, 826, 827, 828, 829, 830, 831, 832, 833, 834, 835, 836, 837, 838, 839, 840, 841,



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|---------------------------------------------------------------------------------------------------------------------------------------------------|------------------|--------------------|--|
| Tester drive carrier for an off-stack IBT-2 module; format S1 (2/3, level 5, S2-S3); schematic only.<br><b>Prototype / schematic + PCB review</b> |                  |                    |  |
| Sheet: /ZLACZA/<br>File: ZLACZA.kicad_sch                                                                                                         |                  |                    |  |
| <b>Title: EGRLab P07 DRIVE (IBT-2 / 2 x BTS7960B)</b>                                                                                             |                  |                    |  |
| Size: A3                                                                                                                                          | Date: 2026-10-05 | <b>Rev: P07-S1</b> |  |
| KiCad E.D.A. 10.0.6                                                                                                                               |                  | Id: 7/8            |  |

P07-S1 / 08 LISTWY SERWISOWE J\_SV1 / J\_SV2 (KRAWEDZ B)

J\_SV2 (S3, x 63,5..96,5) – jedyna listwa (uproszczenie 6.10): 5V\_SYS / 5VA / 3V3A, KPWR, SAFE\_OK, OC\_LOCAL\_N, OC\_GOOD, DRIVE\_OK (1k), GND, I\_T\_OUT (10k), GND.

Rezystor przy weźle (S1 6). Szyna tylko obok GND, innej szyny albo linii logicznej; wezeł analogowy tylko obok GND. Pozostałe wezły: sonda na częściach (ODBIOR).

